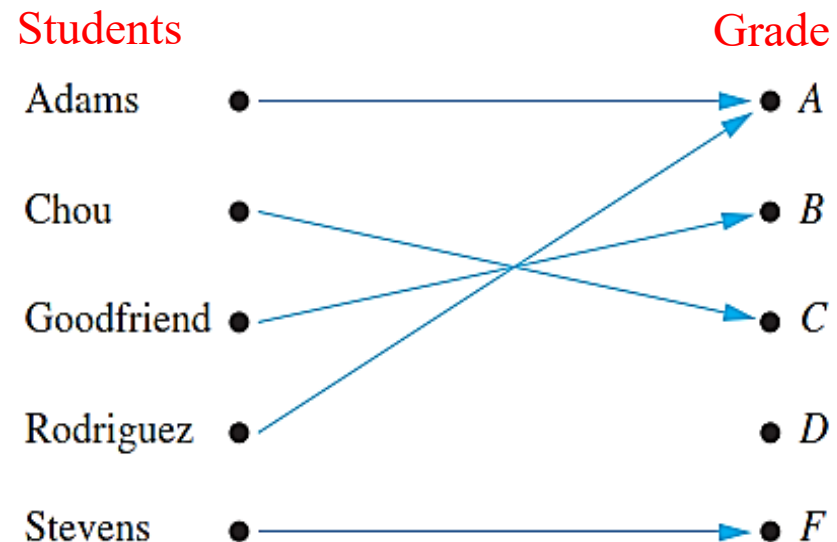


FUNCTIONS

Functions

Definition: Let A and B be nonempty sets. A **function** f from A to B , denoted $f: A \rightarrow B$ is an assignment of each element of A to exactly one element of B . We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A .

Remark: Functions are sometimes also called *mappings* or *transformations*.



Functions

- A function $f: A \rightarrow B$ can also be defined as a subset of $A \times B$ (a relation). This subset is restricted to be a relation where no two elements of the relation have the same first element.
- *Specifically*, a function f from A to B contains one, and only one ordered pair (a, b) for every element $a \in A$.

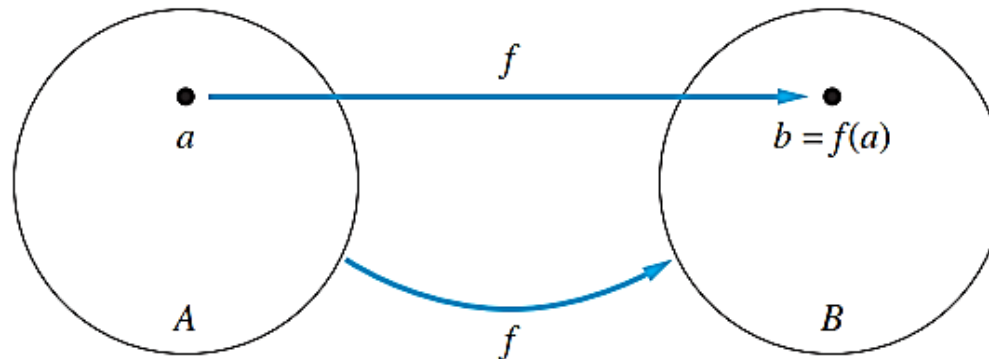
and

$$\forall x [x \in A \rightarrow \exists y [y \in B \wedge (x, y) \in f]]$$

$$\forall x, y_1, y_2 [(x, y_1) \in f \wedge (x, y_2) \in f] \rightarrow y_1 = y_2]$$

Functions

The function f maps A to B



Given a function $f: A \rightarrow B$

- We say f **maps** A to B or f is a **mapping** from A to B .
- A is called the **domain** of f .
- B is called the **codomain** of f . If $f(a) = b$, then
 - b is called the **image** of a under f .
 - a is called the **preimage** of b .

Functions

- The range of f is the set of all *images* of points in A under f and we denote it by $f(A)$.

If f and g are functions from A to B , then they are said to be equal (i.e., $f = g$) if the following conditions hold.

- 1- They have the same domain. i.e., $D(f) = D(g)$
- 2- They have the same codomain. i.e., $R(f) = R(g)$
- 3- They map each element of their common domain to the same element in their common codomain.

$$f(x) = g(x) \quad \forall x \in A$$

Example

Let $A = \{1, 2, 3, 4\}$; $B = \{1, 2, 7, 8, 17, 18, 31, 32\}$ and the function $f: A \rightarrow B$ defined by $f = \{(1, 2), (2, 8), (3, 18), (4, 32)\}$.

Consider another function $g: A \rightarrow \mathbb{N}$ defined by $g(x) = 2x^2$.

Now it is clear that

$D(f) = \{1, 2, 3, 4\}$ with $f(1) = 2, f(2) = 8, f(3) = 18, f(4) = 32$.

Similarly, $D(g) = A = \{1, 2, 3, 4\}$ with $g(1) = 2, g(2) = 8$,
 $g(3) = 18, g(4) = 32$.

Therefore, we get

$$(a) D(f) = \{1, 2, 3, 4\} = D(g)$$

$$(b) R(f) = \{2, 8, 18, 32\} = R(g) \text{ and}$$

$$(c) f(x) = g(x) \forall x \in \{1, 2, 3, 4\}. \text{ This implies that } f = g.$$

Functions

- A function is called real-valued if its codomain is the set of real numbers, and it is called integer-valued if its codomain is the set of integers.
- Two real-valued functions or two integer valued functions with the same domain can be added, as well as multiplied.

Let f_1 and f_2 be functions from A to $\mathbb{R}$. Then $f_1 + f_2$ and $f_1 f_2$ are also functions from A to $\mathbb{R}$ defined for all $x \in A$ by

$$(f_1 + f_2)(x) = f_1(x) + f_2(x)$$

$$(f_1 f_2)(x) = f_1(x) f_2(x)$$

Question

Let f_1 and f_2 be functions from $\mathbb{R}$ to $\mathbb{R}$ such that $f_1(x) = x^2$ and $f_2(x) = x - x^2$.

What are the functions $f_1 + f_2$ and $f_1 f_2$?

Solution: From the definition of the sum and product of functions, it follows that

$$(f_1 + f_2)(x) = f_1(x) + f_2(x) = x^2 + (x - x^2) = x$$

and

$$(f_1 f_2)(x) = x^2(x - x^2) = x^3 - x^4.$$

Injection (one-to-one)

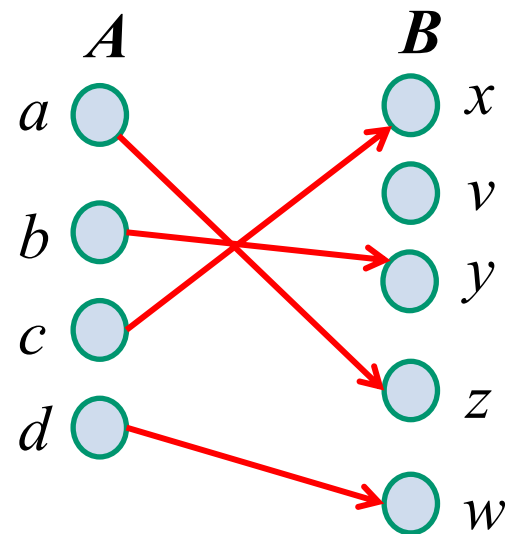
A function f is said to be **one-to-one**, or **an injection**, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f . A function is said to be an **injective** if it is one-to-one.

e.g., The function $f(x) = x - 1$ from $\mathbb{Z}$ to $\mathbb{Z}$ is one-to-one since

$$f(a) = f(b)$$

$$\rightarrow a - 1 = b - 1$$

$$\rightarrow a = b$$



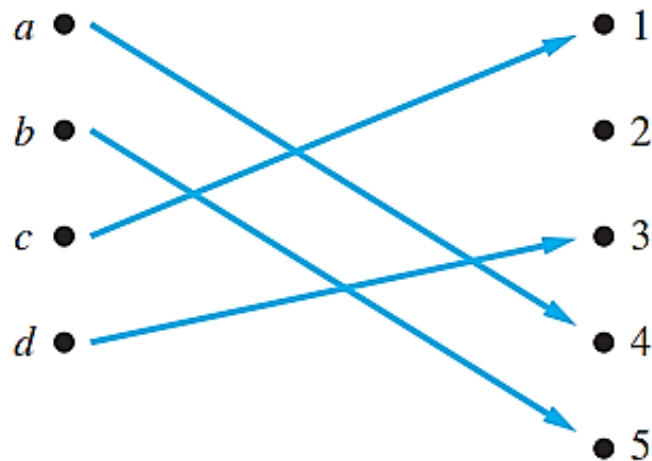
A One-to-One Function

Question

Determine whether the function f from $\{a, b, c, d\}$ to $\{1, 2, 3, 4, 5\}$ with $f(a) = 4$, $f(b) = 5$, $f(c) = 1$, and $f(d) = 3$ is one-to-one.

Answer

The function f is one-to-one because f takes on different values at the four elements of its domain as shown in figure



Question

Determine whether each of these functions from $\{a, b, c, d\}$ to itself is one-to-one.

a) $f(a) = b, f(b) = a, f(c) = c, f(d) = d$

b) $f(a) = b, f(b) = b, f(c) = d, f(d) = c$

c) $f(a) = d, f(b) = b, f(c) = c, f(d) = d$

Answer

a) This is one-to-one.

b) This is **not** one-to-one, since b is the image of both a and b .

c) This is **not** one-to-one, since d is the image of both a and d .

Question

Determine whether the function $f(x) = x^2$ from the set of integers to the set of integers is one-to-one.

Answer

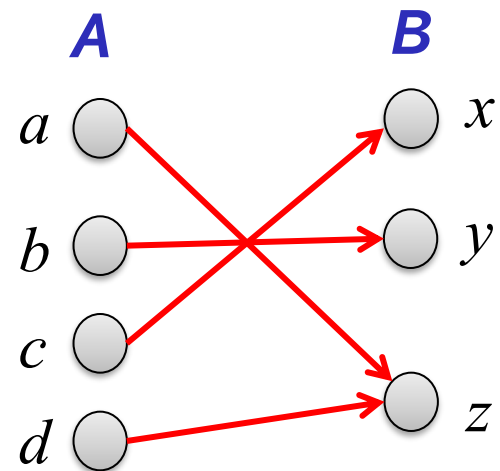
The function $f(x) = x^2$ is not one-to-one because, for instance, $f(1) = f(-1) = 1$, but $1 \neq -1$.

Note that the function $f(x) = x^2$ with its domain restricted to $\mathbb{Z}^+$ is one-to-one.

Surjection (onto)

A function f from A to B is called **onto**, or **surjection**, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$. A function f is called a **surjective** if it is **onto**.

e.g., The function $f(x) = x - 1$ from $\mathbb{Z}$ to $\mathbb{Z}$ is onto since for all $b \in \mathbb{Z}$, there is $a \in \mathbb{Z}$ such that $f(a) = b$ if and only if $a - 1 = b$ which holds if and only if $a = b + 1$.



An Onto Function

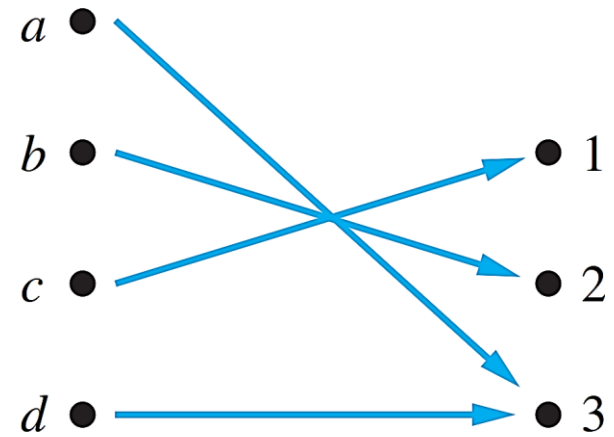
Question

Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3\}$ defined by $f(a) = 3$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$.

Is f an onto function.

Solution

Because all three elements of the codomain are images of elements in the domain, we see that f is onto.



Note that if the codomain were $\{1, 2, 3, 4\}$, then f would not be onto.

Question

Is the function $f(x) = x^2$ from the set of integers to the set of integers ($f: \mathbb{Z} \rightarrow \mathbb{Z}$) onto.

Answer

The function f is not onto because there is no integer x with $x^2 = -1$, for instance.

Question

Determine whether each of these functions from $\{a, b, c, d\}$ to itself is onto.

a) $f(a) = b, f(b) = a, f(c) = c, f(d) = d$

b) $f(a) = b, f(b) = b, f(c) = d, f(d) = c$

c) $f(a) = d, f(b) = b, f(c) = c, f(d) = d$

Answer

a) This is onto because the range is all $\{a, b, c, d\}$.

b) This is **not** onto [The range is $\{b, c\} \neq \{a, b, c, d\}$]

c) This is **not** onto [The range is $\{b, c, d\} \neq \{a, b, c, d\}$]

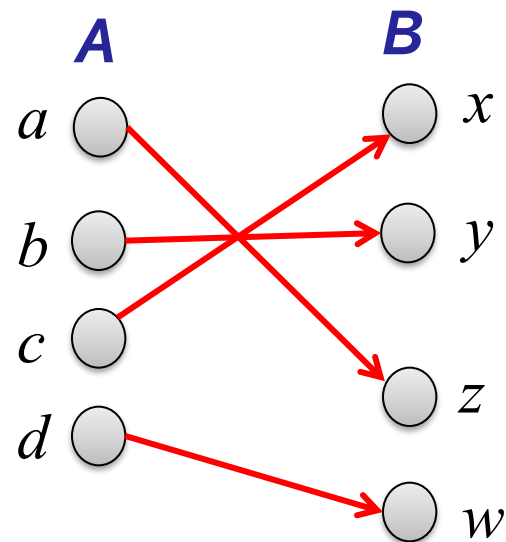
Bijections

(one-to-one correspondence)

A function f is a **one-to-one correspondence**, or a **bijection**, if it is both one-to-one and onto (*surjective* and *injective*).

We also say that such a function is **Bijjective**

e.g., The function $f(x) = x - 1$ from $\mathbb{Z}$ to $\mathbb{Z}$ is one-to-one and onto, hence bijective

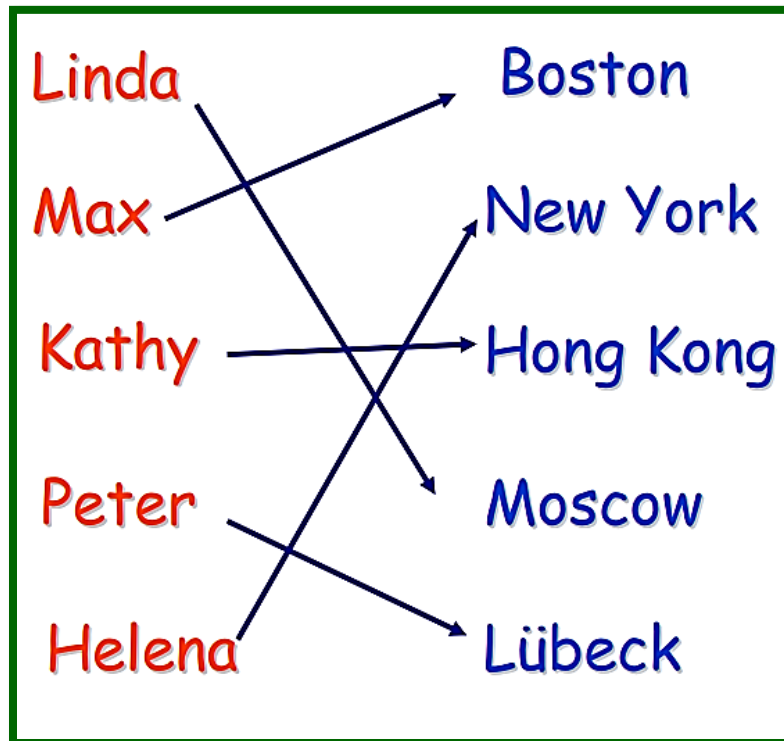


Examples

In the following examples, we use the arrow representation to illustrate functions $f: A \rightarrow B$.

In each example, the complete sets A and B are shown.

(a)



Is f injective?

Yes.

Is f surjective?

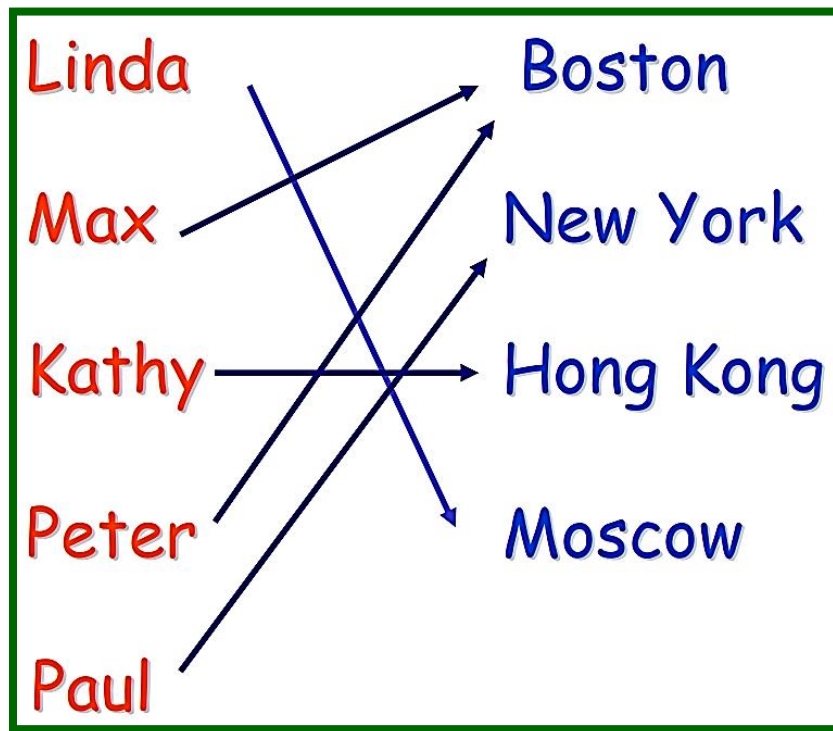
Yes.

Is f bijective?

Yes.

Examples

(b)



Is f injective?

No.

Is f surjective?

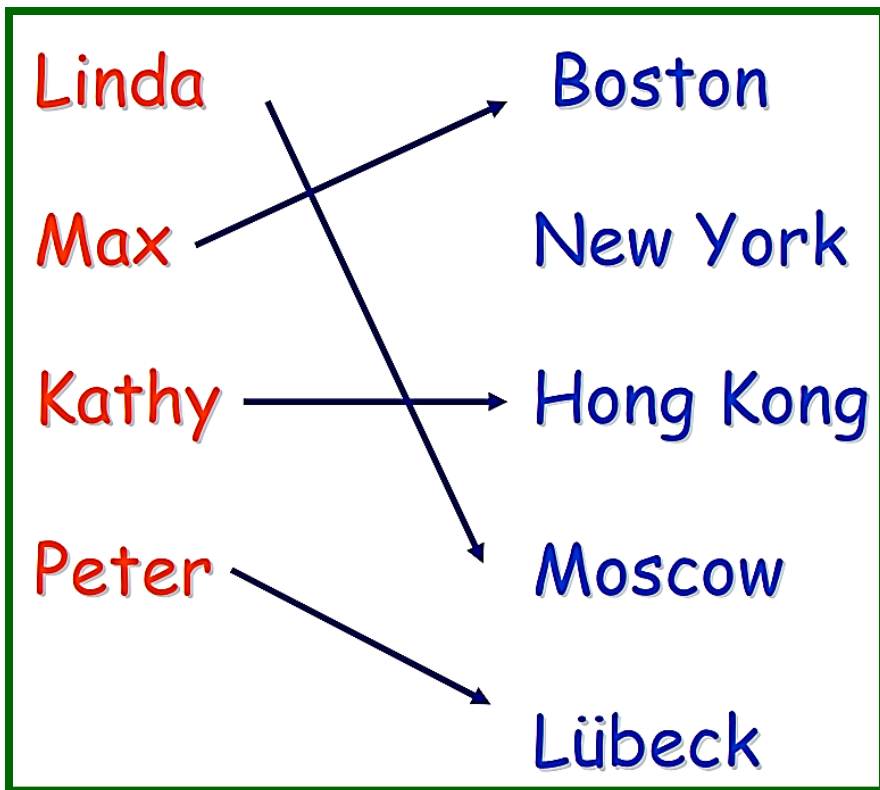
Yes.

Is f bijective?

No.

Examples

(c)



Is f injective?

Yes.

Is f surjective?

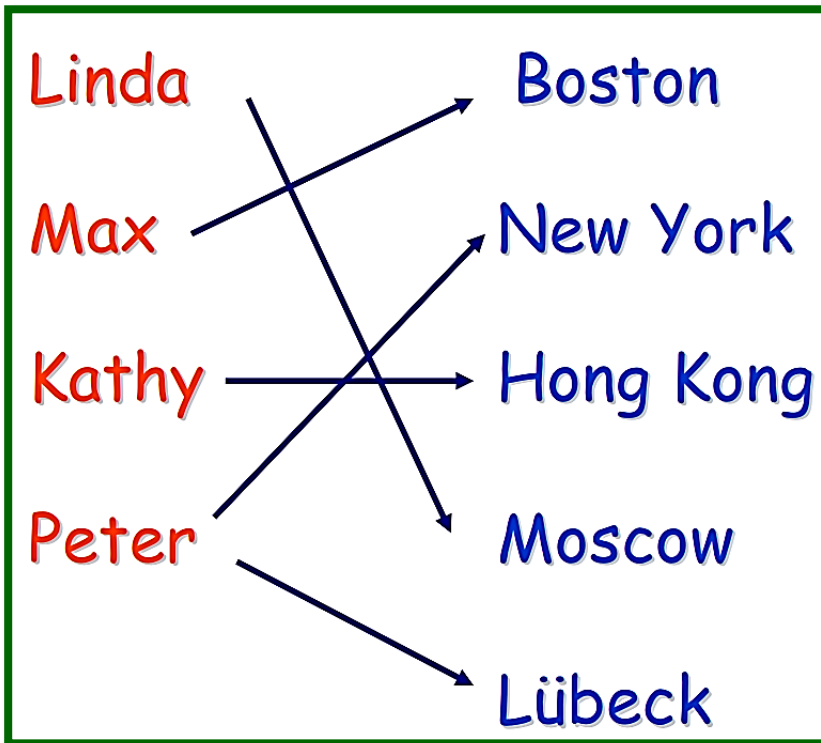
No.

Is f bijective?

No.

Examples

(d)

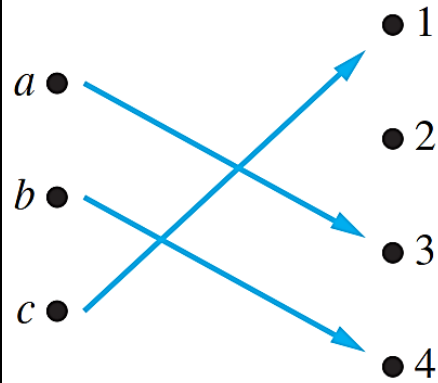


Is f injective?

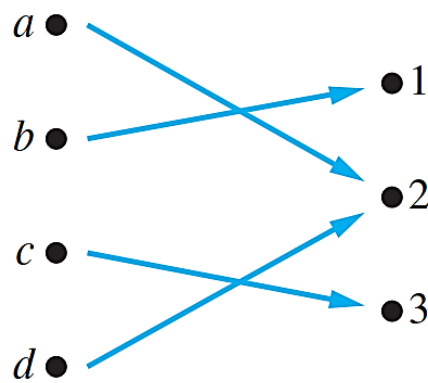
No! f is not even a function!

Examples of Different Types of Correspondences

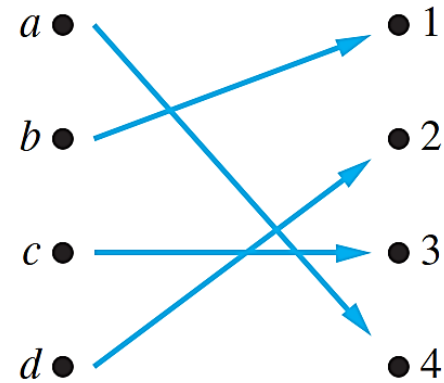
(a) One-to-one,
not onto



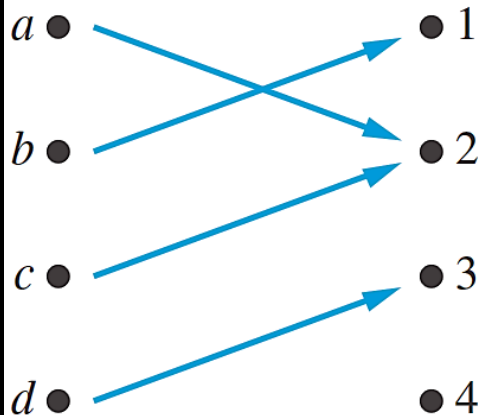
(b) Onto,
not one-to-one



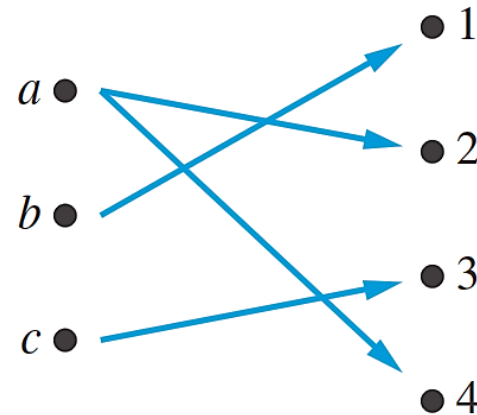
(c) One-to-one,
and onto



(d) Neither one-to-one
nor onto



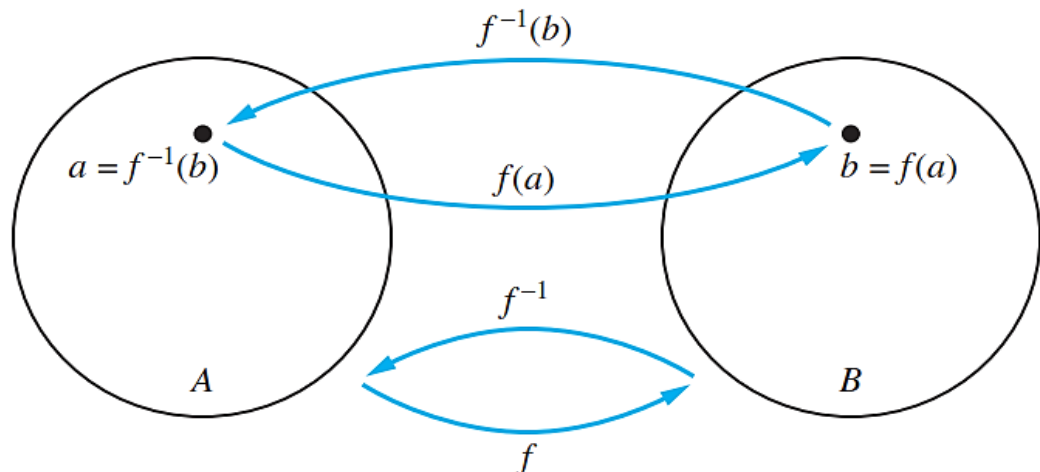
(e) Not a function



Inverse Functions

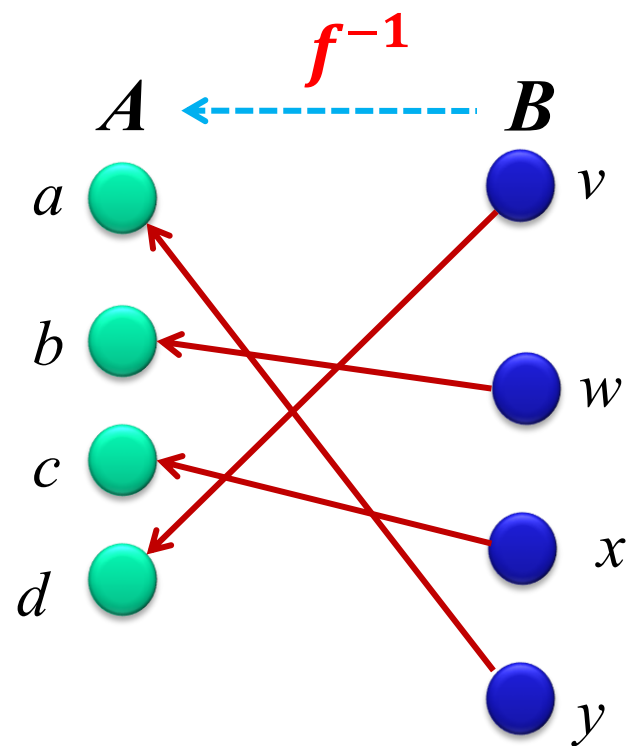
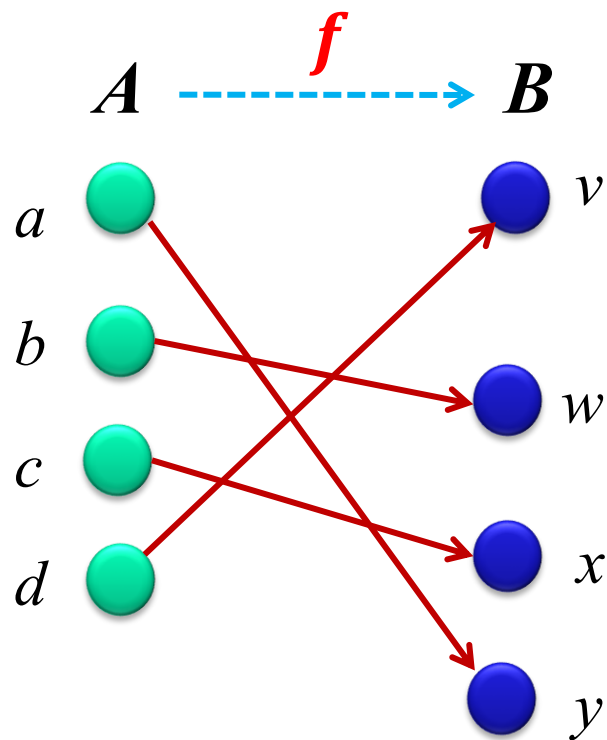
Let f be a one-to-one correspondence from the set A to the set B (f be a bijection from A to B). The inverse function of f is the function that assigns to an element b belonging to B the unique element a in A such that $f(a) = b$.

The inverse function of f is denoted by f^{-1} . Hence,
 $f^{-1}(b) = a$ when $f(a) = b$.



The Function f^{-1} Is the Inverse of Function f .

Inverse Functions



Questions

Let f be the function from $\{a, b, c\}$ to $\{1, 2, 3\}$ such that

$f(a) = 2$, $f(b) = 3$, and $f(c) = 1$. Is f invertible and if so, what is its inverse?

Solution: The function f is invertible because it is a one-to-one correspondence. The inverse function f^{-1} reverses the correspondence given by f , so $f^{-1}(1) = c$, $f^{-1}(2) = a$, and $f^{-1}(3) = b$.

Question

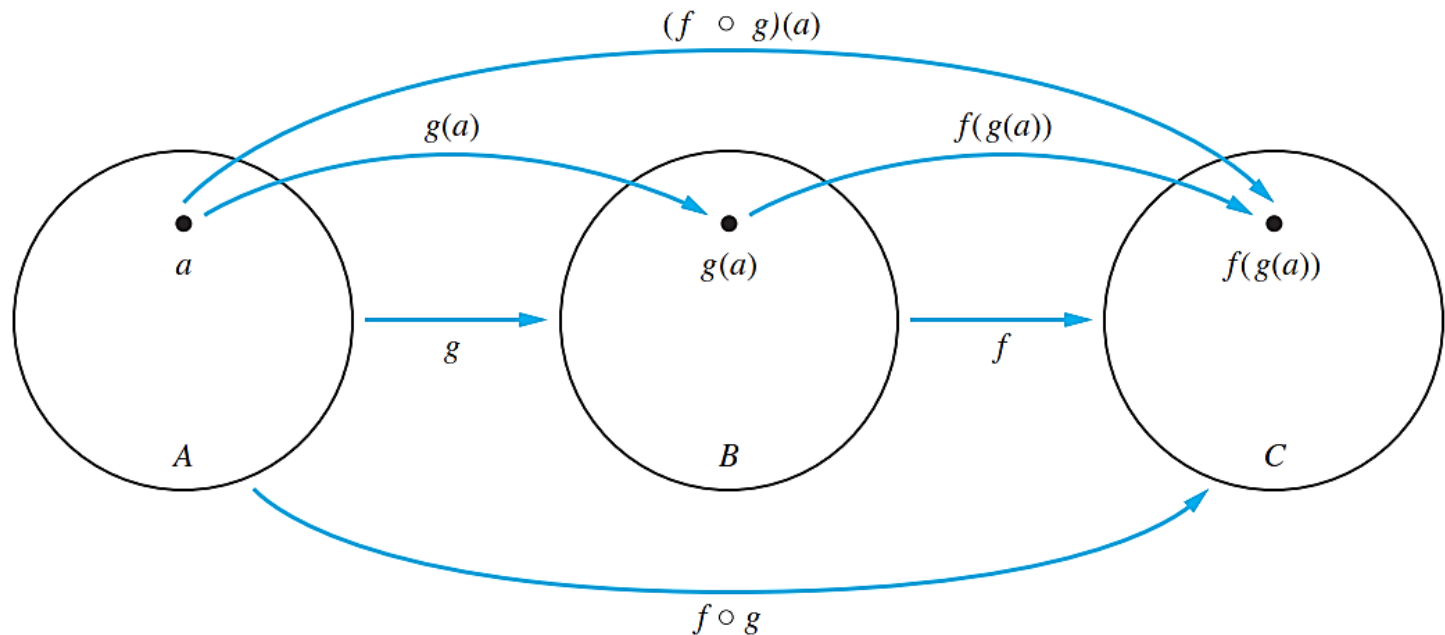
Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ be such that $f(x) = x + 1$. Is f invertible, and if so, what is its inverse?

Solution: The function f is invertible because it is a one-to-one correspondence ($f(a) = f(b) \rightarrow a + 1 = b + 1 \rightarrow a = b$). The inverse function f^{-1} reverses the correspondence so

$$f^{-1}(x) = x - 1.$$

Composition of the Functions

Let g be a function from the set A to the set B and let f be a function from the set B to the set C . The composition of the functions f and g , denoted for all $a \in A$ by $f \circ g$, is defined by $(f \circ g)(a) = f(g(a))$.



Question

Let g be the function from the set $\{a, b, c\}$ to itself such that $g(a) = b$, $g(b) = c$, and $g(c) = a$. Let f be the function from the set $\{a, b, c\}$ to the set $\{1, 2, 3\}$ such that $f(a) = 3$, $f(b) = 2$, and $f(c) = 1$. What is the composition of f and g , and what is the composition of g and f ?

Solution The composition $f \circ g$ is defined by

$$(f \circ g)(a) = f(g(a)) = f(b) = 2,$$

$$(f \circ g)(b) = f(g(b)) = f(c) = 1,$$

$$\text{and } (f \circ g)(c) = f(g(c)) = f(a) = 3.$$

Note that $g \circ f$ is not defined, because the range of f is not a subset of the domain of g .

Question

Given $g = \{(1, a), (2, a), (3, c)\}$, a function from $X = \{1, 2, 3\}$ to $Y = \{a, b, c\}$, and $f = \{(a, y), (b, x), (c, z)\}$, a function from Y to $Z = \{x, y, z\}$, then

- (a) Find the composition function from X to Z .
- (b) Find f^{-1}
- (c) Is g invertible.

Answer

- (a) The function $f \circ g = \{(1, y), (2, y), (3, z)\}$.
- (b) $f^{-1} = \{(y, a), (x, b), (z, c)\}$.
- (c) No, because it is not one to one correspondence.